

Sparse Retrieval Paper Excerpt

Sparse Retrieval With Patch Reranking

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Abstract: We evaluate a patch reranker that improves table-region recall without changing OCR. The main result is a 6.3 point F1 gain on scanned appendices.

1. Method

The parser emits candidate blocks, then applies a patch reranker only to regions marked table-like or figure-like.

Equation 1

$$\text{score} = 0.62 \cdot T + 0.28 \cdot V + 0.10 \cdot R$$

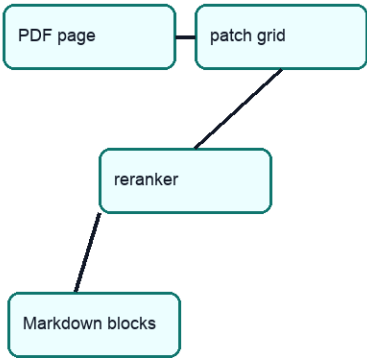
Table 1. Ablation

Variant	Recall	F1
base	71.4	68.2
+patch	80.1	74.5
+caption	82.0	75.1

2. Results

The caption-aware variant is strongest, but most of the gain comes from patch reranking. Failure cases cluster around equation/table boundaries.

Figure 2. Retrieval path



Caption: patch grid feeds the reranker before Markdown block assembly.

3. Limitations

The model still confuses marginal notes with captions when the note touches a figure border. Rotated labels were excluded from the pilot.

Reviewer note

Do not read this box before the Results section; it comments on Figure 2.

Footnote 1: F1 is macro-averaged over pages, not documents.